

E-commerce & Finance Insights Report

Generated via Python ReportLab – Author: Huzeif Khan

Key Performance Indicators (KPIs)

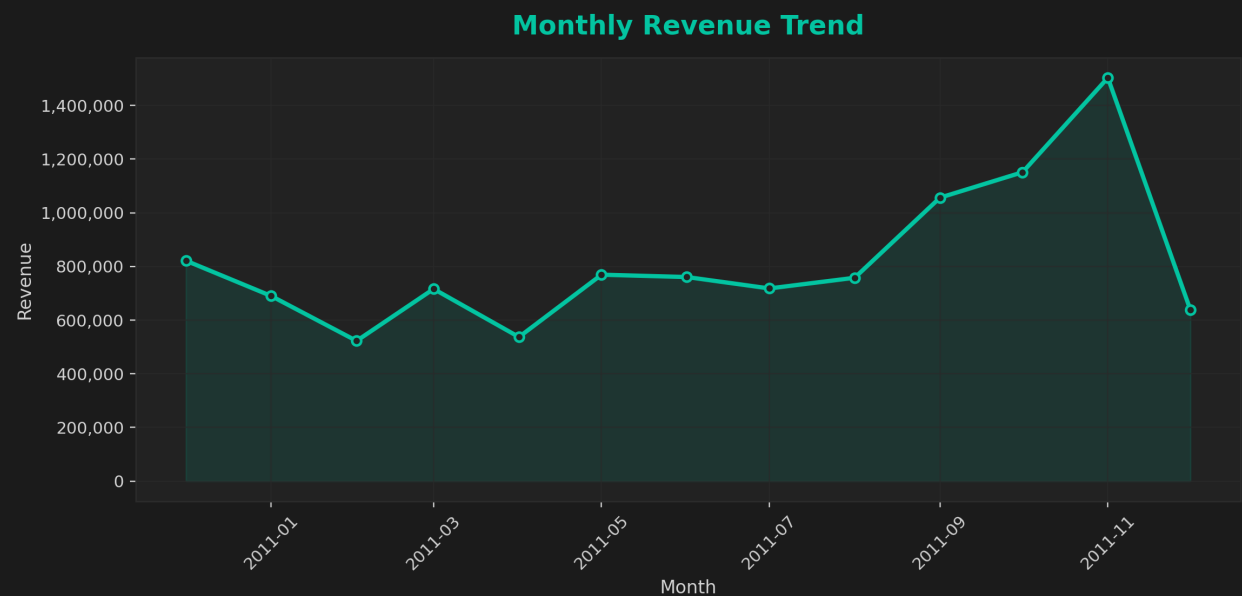
– **Total Revenue:** 10,641,558.95

– **Total Customers:** 4,338

– **Average Order Value:** 2,453.10

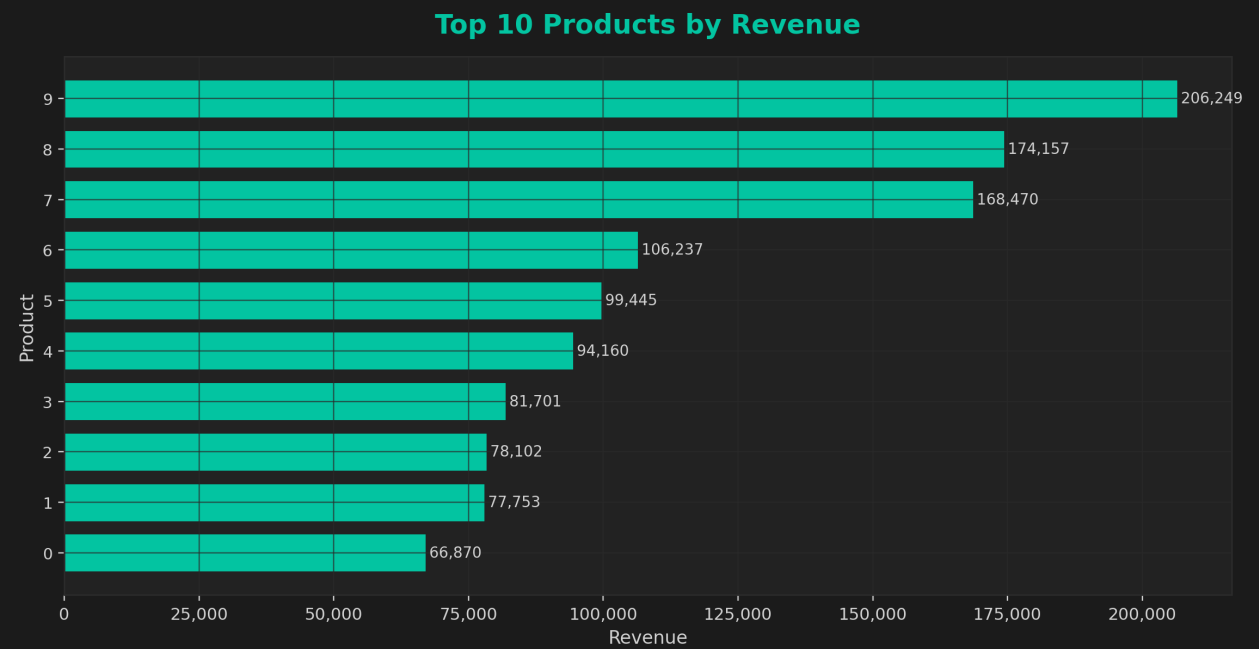
Visual Summary

Monthly Revenue Trend

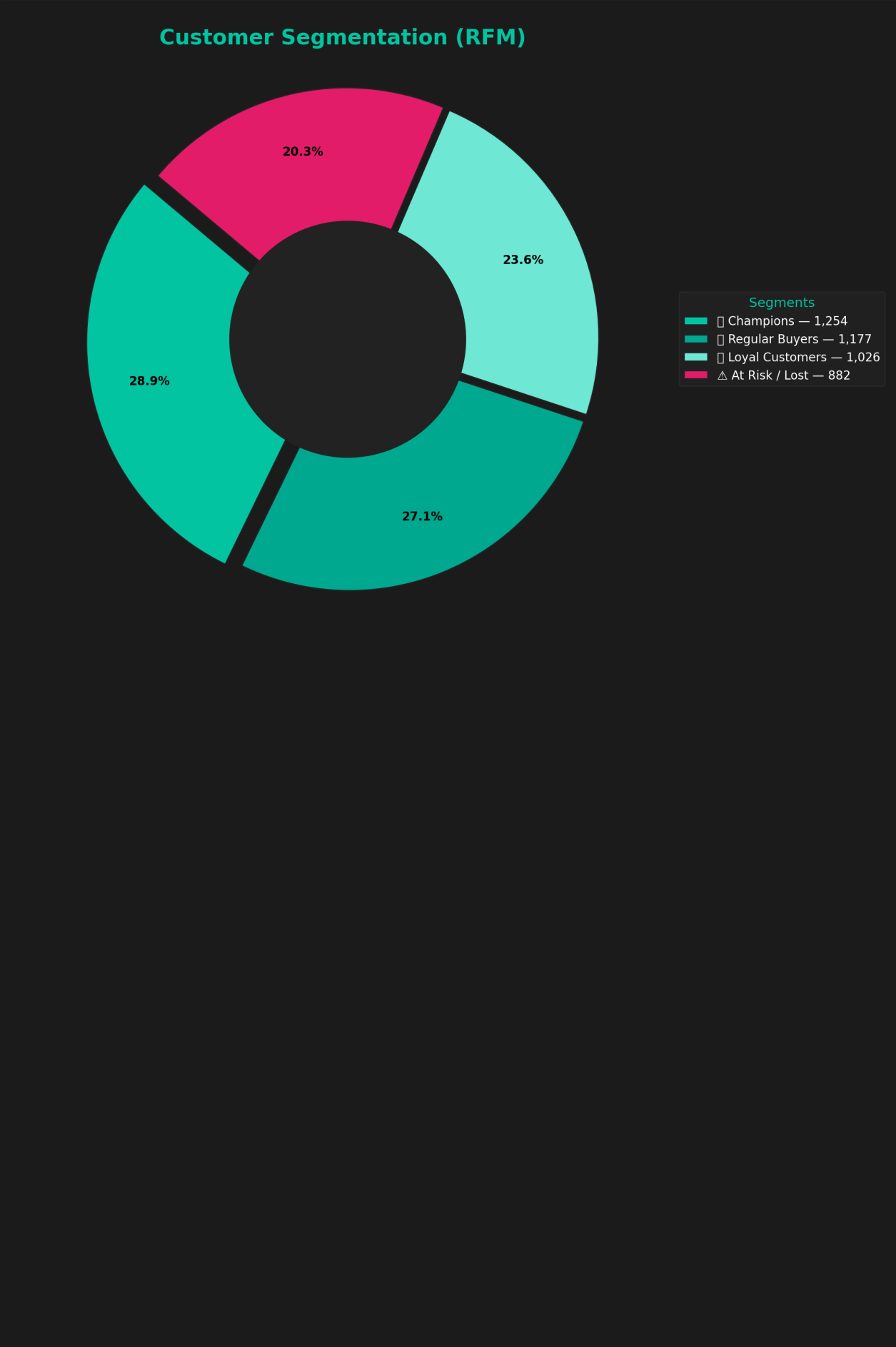


Top 10 Products by Revenue

Top 10 Products by Revenue



Customer Segmentation (RFM Model)



Cohort Retention

Each cell shows the % of customers who returned after N months, grouped by their first purchase month (cohort).



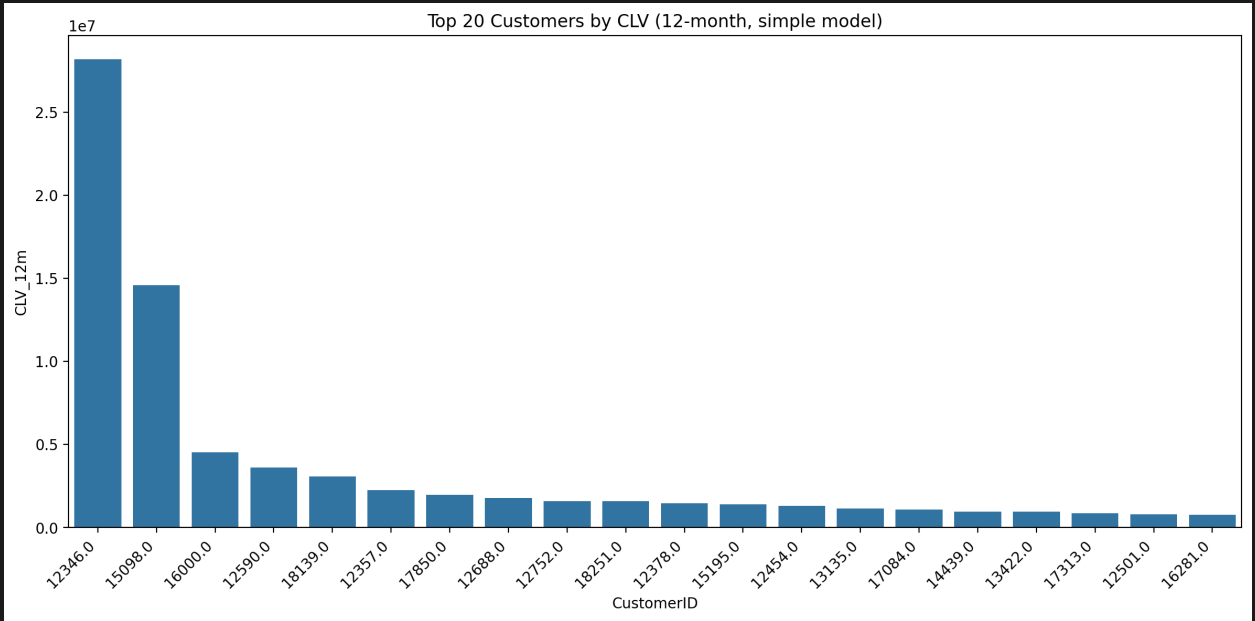
Revenue Forecast (Next 12 Months)

SARIMAX forecast with 80% confidence band. Use this to plan inventory, marketing budgets, and hiring.



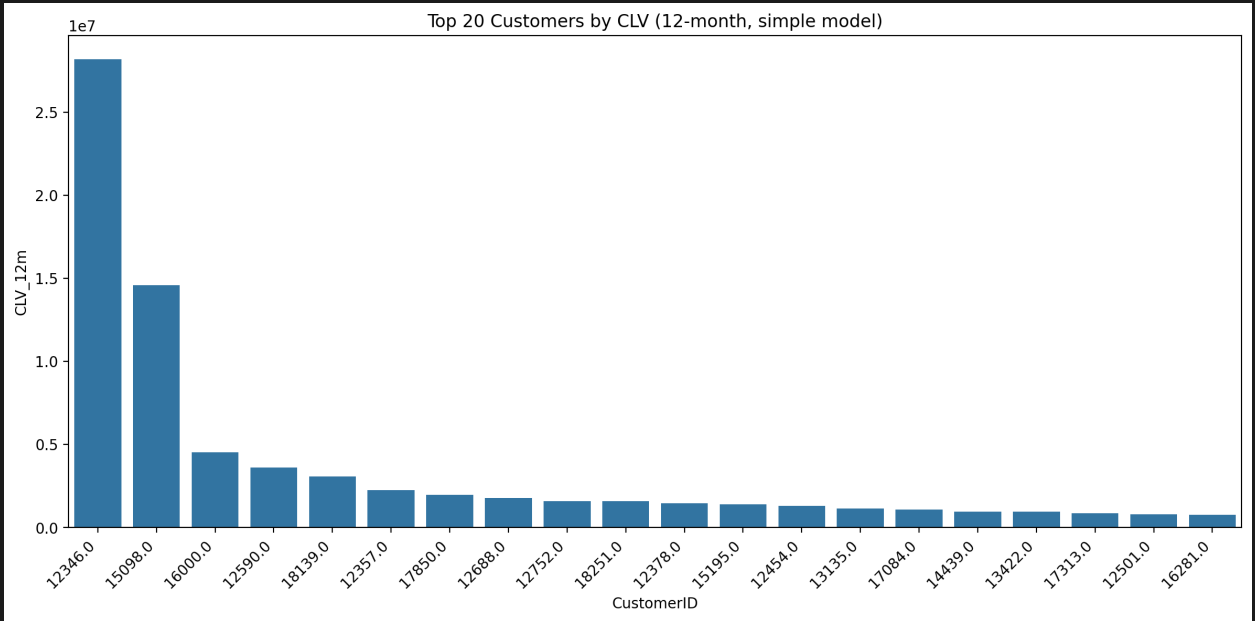
Customer Lifetime Value (CLV) – 6-Month Outlook

We estimate a simple 6-month CLV as $AOV \times (\text{avg monthly repeat probability}) \times 6$, added to the historical revenue. Top 20 customers shown.



Customer Lifetime Value (CLV v1 – 12-Month Model)

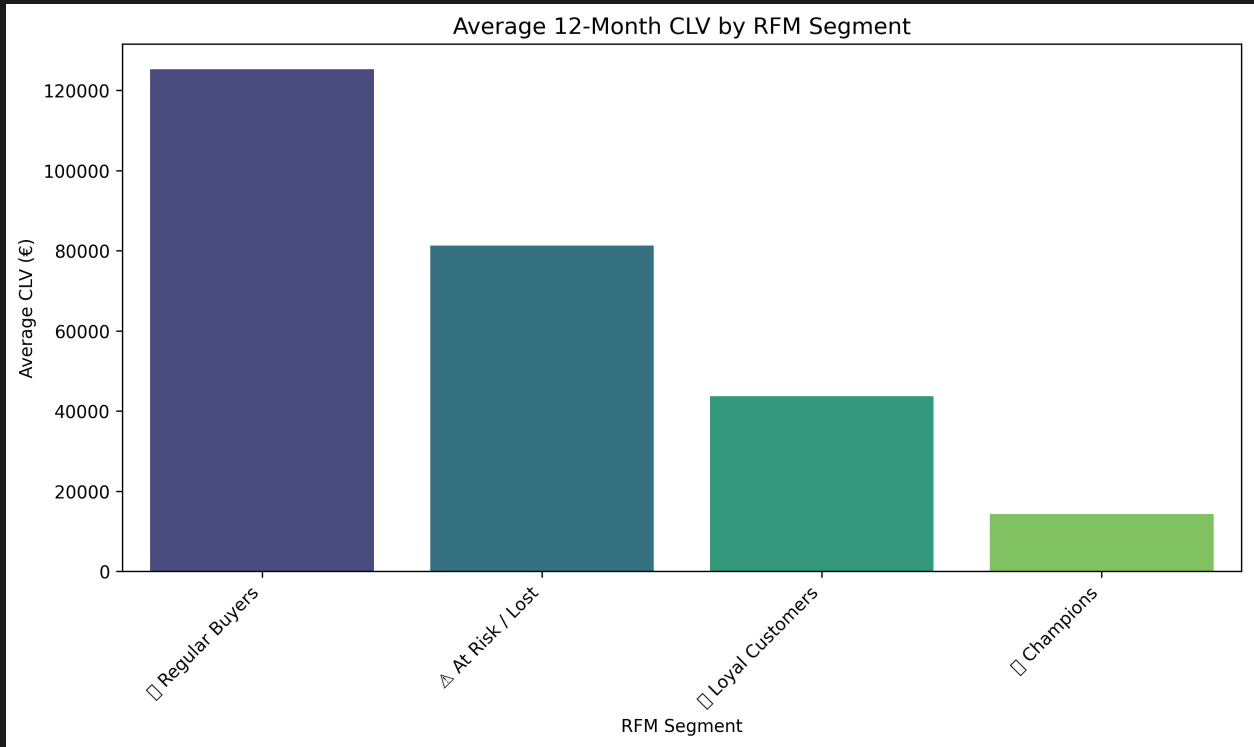
Deterministic CLV v1: AOV – Purchase Frequency per Month – 12 Months. Shows your top-value customers based on average order value and repeat purchase rate.



Last updated (UTC): 2025-11-13 14:41 UTC

CLV by Customer Segment

Average predicted 12-month CLV for each RFM segment.



RFM – CLV Segment Insights

How lifetime value varies by customer segment, and how it correlates with Recency, Frequency and Monetary (RFM).

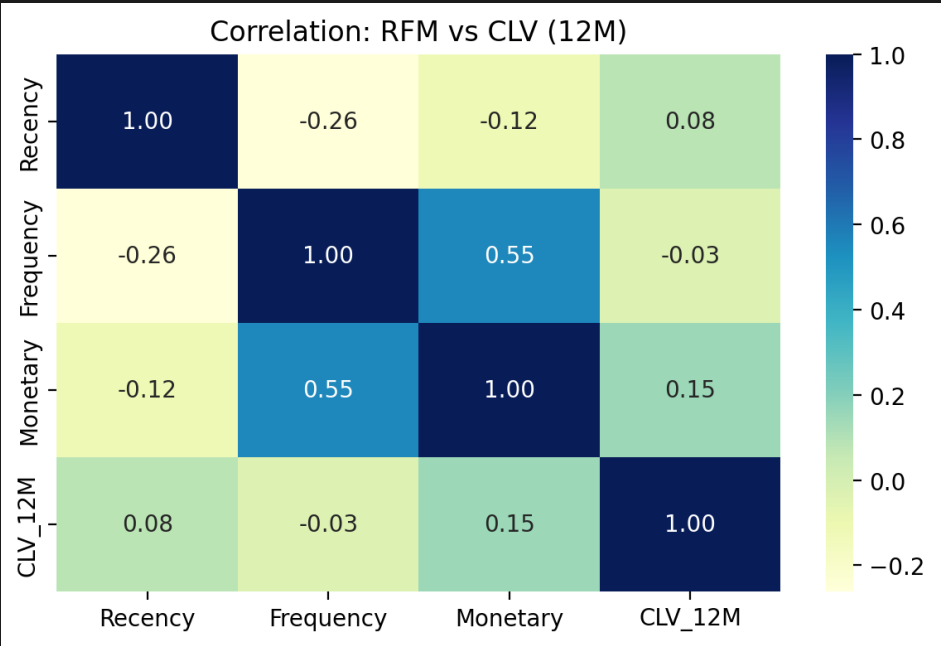
■ Regular Buyers avg CLV \$125,288 (n=1177)

■ At Risk / Lost avg CLV \$81,282 (n=882)

■ Loyal Customers avg CLV \$43,711 (n=1026)

■ Champions avg CLV \$14,341 (n=1253)

RFM–CLV Correlation



CLV vs RFM (example scatter)

